

Companion file — `semantic_layer_demo.py`

1056 lines · sha256 `271b0e57c611` · runs on `pip install duckdb pyyaml`

This is the complete file the article walks through, printed for reading. It is byte-identical to `standalone/semantic_layer_demo.py`, which the project's test suite executes — `tests/test_standalone.py` asserts the numbers it prints against the same warehouse the full pipeline uses.

To run it, copy the file rather than this PDF: a PDF inserts its own line breaks and drops the leading whitespace Python depends on.

```
"""A governed semantic layer for natural-language querying, in one runnable file.
```

```
    pip install duckdb pyyaml
    python semantic_layer_demo.py
```

No API key, no network, no warehouse to provision. It prints five things:

- | | |
|--|--|
| 1. the grain-correct answer | 87.3% (48 / 55), all 7 gates passed |
| 2. the naive answer the same data yields | 88.5% (54 / 61) |
| 3. that same query REFUSED by name | grain_matches |
| 4. why the number moved | argmax over declared process variances |
| 5. four more gates refusing four more | filters_bound, metric_approved, |
| classes of wrong question | time_window_bounded, no_fanout |

THE ARCHITECTURAL RULE, which everything here follows from:

The model never writes SQL and never does arithmetic. It reads a governed metadata slice and emits a validated JSON object -- which metric, which filters, which time window. A deterministic compiler turns that object into SQL.

Text-to-SQL asks a language model to be right about grain, joins, exclusions and fiscal calendars on every call. This asks it to be right about one much easier thing: which approved metric you meant. `resolve\_offline` does that with keyword matching so this file runs with no credentials; `resolve\_with\_claude` is the real thing and is not called.

WHAT IS HONESTLY ELIDED, because one file has a size limit:

- * The data is the June- and July-2026 India slice of the full project -- 83 orders and 87 delivery legs, verbatim. That is the whole population behind 48/55, 54/61 and the June comparison, so every figure is reproduced exactly rather than approximated.
- * `PHASE\_VARIANCE\_CSV` is derived upstream in the full project from per-phase SAP timestamps (`sql/02\_staging\_model.sql`). Here it is embedded. The attribution rule itself -- argmax over declared spans, ties to the earliest phase -- is real code below, and `\_check\_attribution\_covers\_the\_late\_orders` fails loudly if the embedded rows ever stop matching the orders this file derives as late.
- * One metric, one fact table, one grain. The full project has three metrics, nine metadata artifacts, a process model over twelve manufacturing phases, and 401 tests.

Full project: the nine metadata artifacts, the DQ engine, the diagrams and the test suite live in the repository this file was extracted from.

```
"""
```

```
from __future__ import annotations
```

```
import csv
```

```
import io
```

```
import json
```

```
import re
```

```
import sys
```

```
import textwrap
```

```
from dataclasses import dataclass, field
```

```
from datetime import date
```

```
try:
```

```

import duckdb
import yaml
except ImportError:
    sys.exit("pip install duckdb pyyaml")

# pragma: no cover

# The project is frozen at an as-of date so every number is reproducible. A demo that
# says "last month" relative to today is a demo whose output nobody can check.
AS_OF_DATE = date(2026, 8, 7)

# =====
# 1. THE DATA
# =====
# Two tables, and the difference between them is the entire problem. `orders` has one row
# per customer order. `delivery_legs` has one row per PHYSICAL SHIPMENT -- six of these
# orders shipped in two legs, so joining the two and counting rows counts those orders
# twice. If the first leg arrived before the promised date, the order looks on time twice
# over while the leg that actually blew the promise is one row among 61.

ORDERS_CSV = """\
order_id,region,plant,carrier,promised_delivery_date,order_status
SO-1019,IN,1010,BLUEDART,2026-06-13,DLVD
SO-2040,IN,1010,FEDEX,2026-06-05,CANC
SO-2041,IN,1010,BLUEDART,2026-06-08,DLVD
SO-2042,IN,1010,FEDEX,2026-06-24,DLVD
SO-2043,IN,1010,BLUEDART,2026-06-05,DLVD
SO-2044,IN,1010,BLUEDART,2026-06-29,DLVD
SO-2045,IN,1010,FEDEX,2026-06-17,DLVD
SO-2046,IN,1010,BLUEDART,2026-06-29,DLVD
SO-2047,IN,1010,FEDEX,2026-06-01,DLVD
SO-2048,IN,1010,FEDEX,2026-06-01,DLVD
SO-2049,IN,1010,BLUEDART,2026-06-18,DLVD
SO-2050,IN,1010,BLUEDART,2026-06-30,DLVD
SO-2051,IN,1010,FEDEX,2026-06-29,DLVD
SO-2052,IN,1010,BLUEDART,2026-06-29,DLVD
SO-2053,IN,1010,BLUEDART,2026-06-02,DLVD
SO-2054,IN,1010,FEDEX,2026-06-25,DLVD
SO-2055,IN,1010,FEDEX,2026-06-20,DLVD
SO-2056,IN,1010,FEDEX,2026-06-06,DLVD
SO-2057,IN,1010,FEDEX,2026-06-06,DLVD
SO-2058,IN,1010,BLUEDART,2026-06-05,DLVD
SO-2059,IN,1010,FEDEX,2026-06-15,DLVD
SO-2060,IN,1010,BLUEDART,2026-06-30,DLVD
SO-2061,IN,1010,FEDEX,2026-06-11,DLVD
SO-2062,IN,1010,FEDEX,2026-06-20,DLVD
SO-2063,IN,1010,FEDEX,2026-06-15,DLVD
SO-2064,IN,1010,BLUEDART,2026-06-14,DLVD
SO-2065,IN,1010,BLUEDART,2026-06-07,DLVD
SO-1001,IN,1010,BLUEDART,2026-07-13,DLVD
SO-1002,IN,1010,BLUEDART,2026-07-13,DLVD
SO-1003,IN,1010,FEDEX,2026-07-14,DLVD
SO-1004,IN,1010,BLUEDART,2026-07-14,DLVD
SO-1005,IN,1010,BLUEDART,2026-07-15,DLVD
SO-1006,IN,1010,FEDEX,2026-07-15,DLVD

```

SO-1007, IN, 1010, BLUEDART, 2026-07-16, CANC
SO-1008, IN, 1010, BLUEDART, 2026-07-16, DLVD
SO-1009, IN, 1010, FEDEX, 2026-07-17, DLVD
SO-1010, IN, 1010, BLUEDART, 2026-07-18, DLVD
SO-1011, IN, 1010, BLUEDART, 2026-07-24, DLVD
SO-1012, IN, 1010, FEDEX, 2026-07-19, DLVD
SO-1014, IN, 1010, BLUEDART, 2026-07-20, DLVD
SO-1015, IN, 1010, BLUEDART, 2026-07-21, DLVD
SO-1016, IN, 1010, FEDEX, 2026-07-22, DLVD
SO-1017, IN, 1010, BLUEDART, 2026-07-23, DLVD
SO-1018, IN, 1010, BLUEDART, 2026-07-24, DLVD
SO-2001, IN, 1010, FEDEX, 2026-07-02, DLVD
SO-2002, IN, 1010, FEDEX, 2026-07-30, DLVD
SO-2003, IN, 1010, FEDEX, 2026-07-02, DLVD
SO-2004, IN, 1010, FEDEX, 2026-07-27, DLVD
SO-2005, IN, 1010, FEDEX, 2026-07-16, DLVD
SO-2006, IN, 1010, BLUEDART, 2026-07-29, DLVD
SO-2007, IN, 1010, BLUEDART, 2026-07-30, DLVD
SO-2008, IN, 1010, BLUEDART, 2026-07-08, DLVD
SO-2009, IN, 1010, FEDEX, 2026-07-30, DLVD
SO-2010, IN, 1010, FEDEX, 2026-07-02, DLVD
SO-2011, IN, 1010, BLUEDART, 2026-07-18, DLVD
SO-2012, IN, 1010, BLUEDART, 2026-07-04, DLVD
SO-2013, IN, 1010, BLUEDART, 2026-07-28, DLVD
SO-2014, IN, 1010, BLUEDART, 2026-07-24, DLVD
SO-2015, IN, 1010, BLUEDART, 2026-07-15, DLVD
SO-2016, IN, 1010, BLUEDART, 2026-07-07, DLVD
SO-2017, IN, 1010, BLUEDART, 2026-07-29, DLVD
SO-2018, IN, 1010, BLUEDART, 2026-07-31, DLVD
SO-2019, IN, 1010, BLUEDART, 2026-07-11, DLVD
SO-2020, IN, 1010, BLUEDART, 2026-07-06, DLVD
SO-2021, IN, 1010, FEDEX, 2026-07-23, DLVD
SO-2022, IN, 1010, BLUEDART, 2026-07-22, DLVD
SO-2023, IN, 1010, BLUEDART, 2026-07-24, DLVD
SO-2024, IN, 1010, BLUEDART, 2026-07-09, DLVD
SO-2025, IN, 1010, FEDEX, 2026-07-27, DLVD
SO-2026, IN, 1010, FEDEX, 2026-07-25, DLVD
SO-2027, IN, 1010, FEDEX, 2026-07-05, DLVD
SO-2028, IN, 1010, FEDEX, 2026-07-27, DLVD
SO-2029, IN, 1010, FEDEX, 2026-07-25, DLVD
SO-2030, IN, 1010, BLUEDART, 2026-07-09, DLVD
SO-2031, IN, 1010, BLUEDART, 2026-07-07, DLVD
SO-2032, IN, 1010, BLUEDART, 2026-07-08, DLVD
SO-2033, IN, 1010, FEDEX, 2026-07-30, DLVD
SO-2034, IN, 1010, BLUEDART, 2026-07-04, DLVD
SO-2035, IN, 1010, FEDEX, 2026-07-09, DLVD
SO-2036, IN, 1010, BLUEDART, 2026-07-17, DLVD
SO-2037, IN, 1010, BLUEDART, 2026-07-24, DLVD
SO-2038, IN, 1010, BLUEDART, 2026-07-30, DLVD
SO-2039, IN, 1010, BLUEDART, 2026-07-25, DLVD
""

DELIVERY_LEGS_CSV = ""\
delivery_id,order_id,delivery_leg,delivery_date

DL-1019, SO-1019, 1, 2026-06-18
DL-2041, SO-2041, 1, 2026-06-07
DL-2042, SO-2042, 1, 2026-06-23
DL-2043, SO-2043, 1, 2026-06-04
DL-2044, SO-2044, 1, 2026-06-28
DL-2045, SO-2045, 1, 2026-06-16
DL-2046, SO-2046, 1, 2026-06-28
DL-2047, SO-2047, 1, 2026-05-31
DL-2048, SO-2048, 1, 2026-05-31
DL-2049, SO-2049, 1, 2026-06-17
DL-2050, SO-2050, 1, 2026-06-29
DL-2051, SO-2051, 1, 2026-06-28
DL-2052, SO-2052, 1, 2026-06-28
DL-2053, SO-2053, 1, 2026-06-01
DL-2054, SO-2054, 1, 2026-06-24
DL-2055, SO-2055, 1, 2026-06-19
DL-2056, SO-2056, 1, 2026-06-05
DL-2057, SO-2057, 1, 2026-06-05
DL-2058, SO-2058, 1, 2026-06-04
DL-2059, SO-2059, 1, 2026-06-14
DL-2060, SO-2060, 1, 2026-06-29
DL-2061, SO-2061, 1, 2026-06-10
DL-2062, SO-2062, 1, 2026-06-19
DL-2063, SO-2063, 1, 2026-06-14
DL-2064, SO-2064, 1, 2026-06-13
DL-2065, SO-2065, 1, 2026-06-06
DL-1001, SO-1001, 1, 2026-07-12
DL-1002, SO-1002, 1, 2026-07-16
DL-1003, SO-1003, 1, 2026-07-19
DL-1004, SO-1004, 1, 2026-07-18
DL-1005, SO-1005, 1, 2026-07-18
DL-1006, SO-1006, 1, 2026-07-19
DL-1008, SO-1008, 1, 2026-07-15
DL-1009A, SO-1009, 1, 2026-07-16
DL-1009B, SO-1009, 2, 2026-07-20
DL-1010, SO-1010, 1, 2026-07-22
DL-1011, SO-1011, 1, 2026-07-21
DL-1012A, SO-1012, 1, 2026-07-18
DL-1012B, SO-1012, 2, 2026-07-18
DL-1014A, SO-1014, 1, 2026-07-19
DL-1014B, SO-1014, 2, 2026-07-19
DL-1015, SO-1015, 1, 2026-07-19
DL-1016A, SO-1016, 1, 2026-07-21
DL-1016B, SO-1016, 2, 2026-07-21
DL-1017A, SO-1017, 1, 2026-07-22
DL-1017B, SO-1017, 2, 2026-07-22
DL-1018A, SO-1018, 1, 2026-07-23
DL-1018B, SO-1018, 2, 2026-07-23
DL-2001, SO-2001, 1, 2026-07-01
DL-2002, SO-2002, 1, 2026-07-29
DL-2003, SO-2003, 1, 2026-07-01
DL-2004, SO-2004, 1, 2026-07-26
DL-2005, SO-2005, 1, 2026-07-15
DL-2006, SO-2006, 1, 2026-07-28

```
DL-2007,SO-2007,1,2026-07-29
DL-2008,SO-2008,1,2026-07-07
DL-2009,SO-2009,1,2026-07-29
DL-2010,SO-2010,1,2026-07-01
DL-2011,SO-2011,1,2026-07-17
DL-2012,SO-2012,1,2026-07-03
DL-2013,SO-2013,1,2026-07-27
DL-2014,SO-2014,1,2026-07-23
DL-2015,SO-2015,1,2026-07-14
DL-2016,SO-2016,1,2026-07-06
DL-2017,SO-2017,1,2026-07-28
DL-2018,SO-2018,1,2026-07-30
DL-2019,SO-2019,1,2026-07-10
DL-2020,SO-2020,1,2026-07-05
DL-2021,SO-2021,1,2026-07-22
DL-2022,SO-2022,1,2026-07-21
DL-2023,SO-2023,1,2026-07-23
DL-2024,SO-2024,1,2026-07-08
DL-2025,SO-2025,1,2026-07-26
DL-2026,SO-2026,1,2026-07-24
DL-2027,SO-2027,1,2026-07-04
DL-2028,SO-2028,1,2026-07-26
DL-2029,SO-2029,1,2026-07-24
DL-2030,SO-2030,1,2026-07-08
DL-2031,SO-2031,1,2026-07-06
DL-2032,SO-2032,1,2026-07-07
DL-2033,SO-2033,1,2026-07-29
DL-2034,SO-2034,1,2026-07-03
DL-2035,SO-2035,1,2026-07-08
DL-2036,SO-2036,1,2026-07-16
DL-2037,SO-2037,1,2026-07-23
DL-2038,SO-2038,1,2026-07-29
DL-2039,SO-2039,1,2026-07-24
"""
```

```
# variance_days = actual_duration - standard_duration, per manufacturing phase, for the
# orders that missed. Negative means the phase ran UNDER its planned standard. Only late
# orders are attributed: asking "which phase caused the delay" of an on-time order is a
# question with no answer, and inventing one is how dashboards acquire fake precision.
PHASE_VARIANCE_CSV = """\
order_id,mrp,production_order_release,material_staging,production_execution,quality_inspection,goods_rece
ipt,picking,dispatch,transportation,available_slack_days
SO-1019,0,0,0,0,0,0,0,0,5,0
SO-1002,0,0,0,0,4,0,0,0,-1,0
SO-1003,0,0,0,0,0,0,0,0,5,0
SO-1004,0,0,0,0,5,0,0,0,-1,0
SO-1005,0,0,0,4,0,0,0,0,-1,0
SO-1006,0,0,0,0,5,0,0,0,-1,0
SO-1009,0,0,0,0,0,0,0,0,3,0
SO-1010,0,0,0,0,5,0,0,0,-1,0
"""
```

```
# =====
```

2. THE METADATA

```
# =====  
# In the full project this is nine separate governed artifacts with different owners and  
# different review cadences. Collapsed here into one document, but the fields are the  
# real ones, and the field that matters most is `grain`.  
#  
# Read `metrics[0]` and ask: which of these could a crawler pointed at the warehouse  
# have produced? `entity` and `time_dimension`, maybe. Not `grain`, not `exclusions`,  
# not `business_rules`, not `approval_state`. Those are decisions, made by a named human,  
# and they are exactly the fields that decide whether an answer is right rather than  
# merely well-formed.
```

```
SEMANTIC_MODEL_YAML = ""
```

```
entities:
```

- name: fact_order_delivery
 grain: order # one row per order_id -- the load-bearing declaration
 primary_key: order_id
 joins:
 - to: raw_delivery_legs
 on: order_id
 cardinality: one_to_many # this is the hop that turns 87.27% into 88.52%
- name: raw_delivery_legs
 grain: delivery
 primary_key: delivery_id

```
dimensions:
```

- name: region
 entity: fact_order_delivery
 column: region
 allowed_values: [IN, US, EU, APAC] # without this, region='MARS' returns zero rows
- name: plant
 entity: fact_order_delivery
 column: plant
 allowed_values: ["1010", "1710"]
- name: carrier
 entity: fact_order_delivery
 column: carrier
 allowed_values: [BLUEDART, FEDEX, DHL]
- name: delivery_id
 entity: raw_delivery_legs # only reachable through the one_to_many join
 column: delivery_id

```
metrics:
```

- name: on_time_delivery_pct
 label: On-Time Delivery %
 description: >
 Share of eligible customer orders delivered on or before the promised delivery
 date. An order counts as delivered only when its final leg arrives.
 entity: fact_order_delivery
 grain: order
 time_dimension: promised_delivery_date
 expression:
 type: ratio_percent

```

    numerator: "sum(CASE WHEN is_on_time THEN 1 ELSE 0 END)"
    denominator: "sum(CASE WHEN is_eligible THEN 1 ELSE 0 END)"
  exclusions:
    - key: cancelled_orders
      predicate: "order_status <> 'CANC'"
      rationale: A cancelled order was never due; including it inflates the denominator.
    - key: not_yet_due
      predicate: "promised_delivery_date <= DATE '__AS_OF_DATE__'"
      rationale: >
        An order promised after the as-of date is not late, it is pending. Counting it
        as a miss understates performance.
  business_rules:
    - Measured against the CURRENT promised date, not the original.
    - Order-grain only. Delivery-grain aggregation double-counts split shipments.
  owner: VP Supply Chain
  approval_state: approved
  lineage: [SAP SD, SAP EWM, SAP TM, fact_order_delivery]

```

```

- name: production_cycle_time_days
  label: Production Cycle Time (days)
  entity: fact_order_delivery
  grain: order
  time_dimension: promised_delivery_date
  expression: {type: average, numerator: "avg(0)", denominator: "1"}
  approval_state: draft          # deliberately unapproved: the retriever never offers it
  owner: VP Manufacturing

```

Vocabulary. The cheapest phase to build and the one whose absence quietly breaks NL
 # querying: a user who types "OTIF" is asking for a governed metric, and without this
 # the resolver has nothing to map it to.

glossary:

```

- term: On-Time Delivery %
  metric: on_time_delivery_pct
  definition: >
    Share of eligible customer orders delivered on or before the promised delivery
    date. Measured at order grain (one row per order_id). An order counts as delivered
    only when its final delivery leg arrives. Excludes cancelled orders and orders not
    yet due.
  synonyms: [OTD, on time delivery, on-time delivery, delivery performance,
    delivery reliability, promise adherence, OTIF, on time in full]

```

The process model is what makes "why did it drop" answerable. Nine attributable spans
 # with a planned standard each, and ONE attribution rule stated once, in metadata, rather
 # than reinvented per dashboard.

process:

```

  attribution_rule: variance_argmax
  tie_break: earliest_phase_seq
  variances:
    - {phase_seq: 2, key: mrp, label: MRP,
      standard_duration_days: 1}
    - {phase_seq: 3, key: production_order_release, label: Production order release,
      standard_duration_days: 1}
    - {phase_seq: 4, key: material_staging, label: Material staging,
      standard_duration_days: 1}

```

```

- {phase_seq: 5, key: production_execution, label: Production execution,
standard_duration_days: 3}
- {phase_seq: 6, key: quality_inspection, label: Quality inspection,
standard_duration_days: 1}
- {phase_seq: 7, key: goods_receipt, label: Goods receipt,
standard_duration_days: 1}
- {phase_seq: 8, key: picking, label: Picking,
standard_duration_days: 1}
- {phase_seq: 9, key: dispatch, label: Dispatch,
standard_duration_days: 1}
- {phase_seq: 10, key: transportation, label: Transportation,
standard_duration_days: 2}
"""

```

```
def load_model() -> dict:
```

```

    """Parse the governed metadata. The as-of date is substituted at read time, so the
    document itself never carries a date that can silently expire."""

```

```
    text = SEMANTIC_MODEL_YAML.replace("__AS_OF_DATE__", AS_OF_DATE.isoformat())
```

```
    model = yaml.safe_load(text)
```

```
    model["_metrics_by_name"] = {m["name"]: m for m in model["metrics"]}
```

```
    model["_dims_by_name"] = {d["name"]: d for d in model["dimensions"]}
```

```
    model["_entities_by_name"] = {e["name"]: e for e in model["entities"]}
```

```
    return model
```

```
def get_metric(model: dict, name: str) -> dict | None:
```

```
    return model["_metrics_by_name"].get(name)
```

```
# =====
```

```
# 3. THE WAREHOUSE
```

```
# =====
```

```
def build_warehouse() -> duckdb.DuckDBPyConnection:
```

```
    """In-memory DuckDB. The one interesting line is the aggregation to ORDER grain.
```

```

    `final_delivery_date = max(delivery_date)` is what "an order counts as delivered only
    when its final leg arrives" means in SQL. Taking min() -- or joining without
    aggregating at all -- is the 88.5% bug, and it is a one-word difference.

```

```
    """
```

```
    con = duckdb.connect(":memory:")
```

```
    variance_cols = PHASE_VARIANCE_CSV.splitlines()[0].split(",")[1:]
```

```
    for name, ddl, csv_text in (
```

```
        ("raw_orders",
```

```
         "order_id VARCHAR, region VARCHAR, plant VARCHAR, carrier VARCHAR, "
```

```
         "promised_delivery_date DATE, order_status VARCHAR", ORDERS_CSV),
```

```
        ("raw_delivery_legs",
```

```
         "delivery_id VARCHAR, order_id VARCHAR, delivery_leg INTEGER, "
```

```
         "delivery_date DATE", DELIVERY_LEGS_CSV),
```

```
        ("raw_phase_variance",
```

```
         "order_id VARCHAR, " + ", ".join(f"{c} INTEGER" for c in variance_cols),
```

```
         PHASE_VARIANCE_CSV),
```

```

):
    _load(con, name, ddl, csv_text)

con.execute(f"""
CREATE TABLE fact_order_delivery AS
WITH final_leg AS (
    SELECT order_id,
           max(delivery_date) AS final_delivery_date,
           count(*)          AS delivery_count
    FROM raw_delivery_legs
    GROUP BY order_id          -- <-- collapse to one row per ORDER
)
SELECT o.order_id, o.region, o.plant, o.carrier,
       o.promised_delivery_date, o.order_status,
       f.final_delivery_date,
       coalesce(f.delivery_count, 0) AS delivery_count,
       -- The two exclusions, materialised as a single eligibility flag.
       (o.order_status <> 'CANC'
        AND o.promised_delivery_date <= DATE '{AS_OF_DATE}') AS is_eligible,
       (f.final_delivery_date IS NOT NULL
        AND f.final_delivery_date <= o.promised_delivery_date) AS is_on_time
FROM raw_orders o
LEFT JOIN final_leg f USING (order_id)
""")
return con

```

```

def _load(con: duckdb.DuckDBPyConnection, table: str, ddl: str, csv_text: str) -> None:
    """Explicit DDL rather than `read_csv_auto`, so `promised_delivery_date` is a DATE and
    not a string that compares lexicographically and looks like it worked."""
    rows = list(csv.reader(io.StringIO(csv_text)))
    con.execute(f"CREATE TABLE {table} ({ddl})")
    marks = ", ".join("?" * len(rows[0]))
    con.executemany(f"INSERT INTO {table} VALUES ({marks})", rows[1:])

```

```

# =====
# 4. THE INTENT -- the only thing the model is allowed to produce
# =====

```

```

@dataclass(frozen=True)
class Filter:
    column: str
    operator: str      # one of = in between >= <=
    value: object

```

```

@dataclass(frozen=True)
class QueryIntent:
    """What a language model is permitted to emit. Note what is NOT in here: no SQL, no
    column expressions, no arithmetic, no join instructions. Every field is a reference
    to something the metadata already declares, which is what makes it checkable."""
    intent_type: str      # descriptive | diagnostic

```

```

metric: str
grain: str
time_window: tuple[str, str]
time_label: str = ""
dimensions: tuple[str, ...] = ()
filters: tuple[Filter, ...] = ()
compare_to: tuple[str, str] | None = None
compare_label: str = ""

def to_json(self) -> str:
    return json.dumps({
        "intent_type": self.intent_type, "metric": self.metric, "grain": self.grain,
        "time_window": list(self.time_window), "dimensions": list(self.dimensions),
        "filters": [{"column": f.column, "operator": f.operator, "value": f.value}
                     for f in self.filters],
    }, indent=2)

```

```

# =====
# 5. THE RESOLVER -- the only stage that calls a model
# =====

```

```

def resolve_offline(question: str, model: dict) -> QueryIntent:
    """Keyword resolution, so this file runs with no credentials.

```

This is a stand-in for `resolve\_with\_claude` below, and the substitution is the point: every stage downstream receives a `QueryIntent` and cannot tell which produced it. If swapping the resolver changed anything about how the answer is computed, the architecture would be leaking.

```

    """
    q = question.lower()

```

```

    metric = None
    for term in model["glossary"]:
        candidates = [term["term"]] + term["synonyms"]
        if any(c.lower() in q for c in candidates):
            metric = term["metric"]
            break
    if metric is None:
        raise ValueError(
            f"No governed metric matches {question!r}. The glossary knows: "
            + ", ".join(sorted({s for t in model['glossary'] for s in t['synonyms']})))
    )

```

```

# Two traps here, both of which this code originally fell into, and both of which are
# about the DIMENSION VALUES rather than about the metric:
#

```

```

#   substring: "IN" occurs inside "inspection"
#   stopword:  "IN" IS an English word -- "delivery reliability for FEDEX in APAC"
#             resolves to region=IN if the code is matched case-insensitively
#

```

```

# So the region CODE is matched case-sensitively against the original question, and
# the human words are matched case-insensitively as declared aliases. Note that both

```

```

# bugs produce a perfectly LEGAL filter, which every gate downstream passes -- the
# gates protect the arithmetic, not the reading comprehension. That asymmetry is the
# reason the real resolver is a model handed `allowed_values`, and the reason it is
# confined to choosing rather than computing.
aliases = {"IN": ["india", "indian"], "EU": ["europe", "european", "emea"],
          "US": ["usa", "united states", "north america"],
          "APAC": ["asia pacific", "asia-pacific"]}
filters = []
for dim in ("region", "plant", "carrier"):
    for allowed in model["_dims_by_name"][dim]["allowed_values"]:
        hit = re.search(rf"\b{re.escape(allowed)}\b", question) or any(
            re.search(rf"\b{re.escape(a)}\b", q) for a in aliases.get(allowed, []))
        if hit:
            filters.append(Filter(dim, "=", allowed))
            break

diagnostic = any(w in q for w in ("why", "cause", "reason", "drop", "driver"))
return QueryIntent(
    intent_type="diagnostic" if diagnostic else "descriptive",
    metric=metric,
    grain=get_metric(model, metric)["grain"], # from METADATA, never from the text
    time_window=("2026-07-01", "2026-07-31"),
    time_label="July 2026",
    filters=tuple(filters),
    compare_to=("2026-06-01", "2026-06-30") if diagnostic else None,
    compare_label="June 2026" if diagnostic else "",
)

```

```

RESOLVER_SYSTEM_PROMPT = """\

```

```

You select a governed metric. You do not write SQL and you do not do arithmetic.

```

```

Return ONLY a QueryIntent. Every value must come from the metadata slice provided:

```

- metric: an exact name from the approved metric list
- grain: the metric's declared grain, copied verbatim, never inferred from the question
- filters: only columns listed as dimensions, only values from allowed_values
- time_window: explicit start and end dates, never open-ended

```

If the question cannot be answered from this slice, say so instead of approximating.

```

```

An approximation that looks like an answer is worse than a refusal.

```

```

"""

```

```

def resolve_with_claude(question: str, model: dict) -> QueryIntent: # pragma: no cover
    """The real resolver. Not called by `main` -- shown because it is the whole point.

```

```

Two things make this safe, and neither is prompt wording. First, the retriever hands
over only APPROVED metrics, so an unapproved one cannot be picked: it was never
offered. Second, whatever comes back goes through all seven gates before it can
become SQL, so a wrong intent is refused rather than answered.

```

```

"""

```

```

import anthropic

```

```

approved = [m for m in model["metrics"] if m["approval_state"] == "approved"]

```

```

slice_for_model = {
    "metrics": [{k: m[k] for k in ("name", "label", "grain", "description")
                    if k in m} for m in approved],
    "dimensions": [{"name": d["name"], "allowed_values": d.get("allowed_values")}
                    for d in model["dimensions"]],
    "glossary": [{"term": t["term"], "metric": t["metric"],
                    "synonyms": t["synonyms"]} for t in model["glossary"]],
}

tool = {
    "name": "emit_query_intent",
    "description": "Emit the governed query intent for this question.",
    "input_schema": {
        "type": "object",
        "properties": {
            "intent_type": {"enum": ["descriptive", "diagnostic"]},
            "metric": {"enum": [m["name"] for m in approved]},
            "grain": {"type": "string"},
            "time_window": {"type": "array", "items": {"type": "string"},
                            "minItems": 2, "maxItems": 2},
            "dimensions": {"type": "array", "items": {"type": "string"}},
            "filters": {"type": "array", "items": {
                "type": "object",
                "properties": {"column": {"type": "string"},
                              "operator": {"enum": ["=", "in", "between",
                                                    ">=", "<="]},
                              "value": {}}},
                "required": ["column", "operator", "value"]}},
        },
        "required": ["intent_type", "metric", "grain", "time_window"],
    },
}

reply = anthropic.Anthropic().messages.create(
    model="claude-sonnet-5",
    max_tokens=1024,
    system=RESOLVER_SYSTEM_PROMPT,
    tools=[tool],
    tool_choice={"type": "tool", "name": "emit_query_intent"},
    messages=[{"role": "user", "content":
        f"Metadata slice:\n{json.dumps(slice_for_model, indent=2)}\n\n"
        f"Question: {question}"}],
)

args = next(b.input for b in reply.content if b.type == "tool_use")
return QueryIntent(
    intent_type=args["intent_type"], metric=args["metric"], grain=args["grain"],
    time_window=tuple(args["time_window"]),
    dimensions=tuple(args.get("dimensions", ())),
    filters=tuple(Filter(f["column"], f["operator"], f["value"])
                    for f in args.get("filters", ())),
)

```

```

# =====
# 6. THE VALIDATOR -- seven gates, between the model's JSON and any SQL
# =====

```

```

# Three rules that are not negotiable:
#
# * All seven gates always run. A user deserves every reason at once, not the first.
# * A gate reports pass only when it actually verified its property. A gate that could
#   not be evaluated FAILS CLOSED -- "I did not check" must never be recorded as
#   "I checked and it was fine".
# * Validation runs BEFORE compilation, so a refused question never becomes SQL.

GATE_NAMES = ("metric_approved", "dimensions_declared", "filters_bound", "grain_matches",
              "no_fanout", "time_window_bounded", "exclusions_applied")

@dataclass
class ValidationResult:
    passed: dict[str, bool] = field(default_factory=dict)
    failures: list[tuple[str, str]] = field(default_factory=list)

    @property
    def ok(self) -> bool:
        return len(self.passed) == len(GATE_NAMES) and all(self.passed.values())

def validate(qi: QueryIntent, model: dict) -> ValidationResult:
    """Never raises. A hallucinated metric name is a normal, expected input."""
    r = ValidationResult()
    metric = get_metric(model, qi.metric)

    def gate(name: str, ok: bool, why: str = "") -> None:
        r.passed[name] = ok
        if not ok:
            r.failures.append((name, why))

    gate("metric_approved",
        metric is not None and metric.get("approval_state") == "approved",
        f"Metric {qi.metric!r} is not an approved metric. Approved: "
        + ", ".join(m["name"] for m in model["metrics"]
                    if m["approval_state"] == "approved"))

    undeclared = [d for d in qi.dimensions if d not in model["_dims_by_name"]]
    gate("dimensions_declared", not undeclared,
        f"Dimensions {undeclared} are not declared in the semantic model.")

    unbound = [f.column for f in qi.filters if f.column not in model["_dims_by_name"]]
    outside = [f"{f.column}={f.value!r}" for f in qi.filters
               if f.column in model["_dims_by_name"]
               and model["_dims_by_name"][f.column].get("allowed_values")
               and f.value not in model["_dims_by_name"][f.column]["allowed_values"]]
    gate("filters_bound", not unbound and not outside,
        f"Filter columns {unbound} do not bind to a declared dimension."
        if unbound else
        f"Filter {'', '.join(outside)} is outside the declared allowed_values. Without "
        f"this gate the query returns zero rows, and zero rows gets read as an answer.")

    if metric is None:
        gate("grain_matches", False,

```

```

        f"Cannot check grain: metric {qi.metric!r} is not in the semantic model, so "
        f"there is no certified grain to compare {qi.grain!r} against.")
    else:
        gate("grain_matches", qi.grain == metric["grain"],
            f"Metric {qi.metric!r} is certified at grain {metric['grain']!r}; the intent "
            f"asks for grain {qi.grain!r}. Aggregating at the wrong grain changes the "
            f"number.")

    fanout = []
    if metric is not None:
        home = metric["entity"]
        for name in list(qi.dimensions) + [f.column for f in qi.filters]:
            dim = model["_dims_by_name"].get(name)
            if dim is None or dim["entity"] == home:
                continue
            join = next((j for j in model["_entities_by_name"][home].get("joins", []))
                        if j["to"] == dim["entity"]), None)
            # A many_to_one hop cannot duplicate rows and must pass. Only one_to_many
            # fans out. A gate that blocked every join would look correct and quietly
            # break every legitimate cross-entity question.
            if join is None or join["cardinality"] == "one_to_many":
                fanout.append(f"{name} (on {dim['entity']})")
    gate("no_fanout", not fanout,
        f"Reaching {'', '.join(fanout)} requires a one_to_many join, which duplicates "
        f"rows before aggregation.")

    start, end = qi.time_window
    gate("time_window_bounded", bool(start and end) and start <= end,
        f"Time window {qi.time_window} is unbounded or inverted.")

    gate("exclusions_applied",
        metric is not None and all(e.get("predicate") for e in
                                   metric.get("exclusions", [])),
        f"Metric {qi.metric!r} declares an exclusion with no predicate, so it cannot be "
        f"applied. A metric whose exclusions cannot be enforced must not be answered.")
    return r

```

```

# =====
# 7. THE COMPILER -- the only thing that writes SQL
# =====

```

```

def compile_sql(qi: QueryIntent, model: dict,
                window: tuple[str, str] | None = None) -> str:
    """Deterministic. Same intent in, same SQL out, byte for byte.

```

Ratios are compiled as ``sum(numerator) / sum(denominator)``, never as an average of per-group rates. Averaging rates across groups of unequal size gives a different number -- Simpson's paradox -- and no amount of prompting reliably prevents a model from doing it. Here it is structurally impossible.

```

"""

```

```

metric = get_metric(model, qi.metric)
start, end = window or qi.time_window

```

```

expr = metric["expression"]

where = [f"{metric['time_dimension']} BETWEEN DATE '{start}' AND DATE '{end}'"]
where += [e["predicate"] for e in metric.get("exclusions", [])] # unconditional
for f in qi.filters:
    col = model["_dims_by_name"][f.column]["column"]
    where.append(f"{col} {f.operator} '{f.value}'")

group = [model["_dims_by_name"][d]["column"] for d in qi.dimensions]
select = group + [
    f"round(100.0 * {expr['numerator']} / nullif({expr['denominator']}, 0), 4) ",
    f"AS value",
    f"{expr['numerator']} AS numerator",
    f"{expr['denominator']} AS denominator",
]
sql = (f"SELECT {' , '.join(select)}\nFROM {metric['entity']}\n"
      f"WHERE {(chr(10) + ' AND ').join(where)}")
if group:
    sql += f"\nGROUP BY {' , '.join(group)}\nORDER BY {' , '.join(group)}"
return sql

```

```

# =====
# 8. EXECUTOR + PROVENANCE
# =====

```

```

@dataclass
class Answer:
    value: float
    numerator: int
    denominator: int
    sql: str
    intent: QueryIntent
    metric: dict
    gates: dict[str, bool]

    def render(self) -> str:
        m, qi = self.metric, self.intent
        filters = ", ".join(f"{f.column} {f.operator} {f.value}"
                           for f in qi.filters) or "none"
        headline = textwrap.wrap(
            f"{m['label']} is {self.value:.1f}% -- {self.numerator} of "
            f"{self.denominator} eligible orders delivered on or before the promised "
            f"date", 71)
        return "\n".join([
            f"ANSWER {headline[0]}",
            *(f" {ln}" for ln in headline[1:]),
            "",
            f" metric {m['name']}",
            f" arithmetic {self.numerator} / {self.denominator} = "
            f"{100 * self.numerator / self.denominator:.4f}%",
            f" grain {m['grain']} (one row per {m['grain']})",
            f" window {qi.time_label or '-'}"

```

```

        f"({qi.time_window[0]} to {qi.time_window[1]}) on {m['time_dimension']}",
        f" filters      {filters}",
        f" exclusions    {'', '.join(e['key'] for e in m.get('exclusions', []))}",
        f" lineage        {'' -> '.join(m['lineage'])}",
        f" trust          TRUSTED",
        f" gates            {sum(self.gates.values())}/{len(self.gates)} passed: "
        + ", ".join(sorted(self.gates)),
    ])

```

```

def execute(con, qi: QueryIntent, model: dict,
            window: tuple[str, str] | None = None) -> Answer:
    """An empty window yields NULLs, not zeros. Zero is a measurement; NULL is the
    absence of one, and coercing it to 0.0% would report a catastrophe that never
    happened."""
    sql = compile_sql(qi, model, window)
    value, num, den = con.execute(sql).fetchone()
    return Answer(float(value) if value is not None else float("nan"),
                  int(num or 0), int(den or 0), sql, qi,
                  get_metric(model, qi.metric), {})

def answer(con, qi: QueryIntent, model: dict) -> tuple[int, str]:
    """Validate, then compile. Never the other way round.

    The refusal is the feature. "I cannot answer that, and here is which rule stopped me"
    is a system you can put in front of a CFO. One that always produces a number is not.
    """
    result = validate(qi, model)
    if not result.ok:
        lines = [f"REFUSED {qi.metric}: the intent did not pass validation.", "",
                 " gates failed " + ", ".join(g for g, ok in sorted(result.passed.items())
                                                if not ok)]

        for gate_name, why in result.failures:
            lines += _wrap(gate_name, why)
        lines += ["",
                 " No number is shown. A value that failed a gate is not a caveated",
                 " answer, it is an unverified one."]
        return 1, "\n".join(lines)

    ans = execute(con, qi, model)
    ans.gates = result.passed
    out = [ans.render()]
    if qi.intent_type == "diagnostic":
        out.append(_diagnose(con, qi, model, ans))
    return 0, "\n\n".join(out)

```

```

def _wrap(label: str, text: str, width: int = 78) -> list[str]:
    body = textwrap.wrap(text, width - 23)
    return [f" {label:<20} {body[0]}" + [f" {'':<20} {ln}" for ln in body[1:]]

```

```

# =====

```

```

# 9. THE DIAGNOSTIC -- the question BI tools cannot take
# =====
# "What was it?" needs a glossary, a grain and a metric. "WHY did it drop?" needs
# per-phase baselines and an attribution rule declared once, in metadata. This is why
# the manufacturing process is in scope at all: the answer to "why" lives upstream in
# production, not in the delivery table.

def _diagnose(con, qi: QueryIntent, model: dict, ans: Answer) -> str:
    spans = model["process"]["variances"]
    _check_attribution_covers_the_late_orders(con, qi, model)

    cur = con.execute(f"""
        SELECT v.* FROM raw_phase_variance v
        JOIN fact_order_delivery f USING (order_id)
        WHERE f.is_eligible AND NOT f.is_on_time
              AND f.promised_delivery_date BETWEEN DATE '{qi.time_window[0]}'
              AND DATE '{qi.time_window[1]}'
    """)
    names = [d[0] for d in cur.description]
    rows = [dict(zip(names, r)) for r in cur.fetchall()]

    # The attribution rule, applied. argmax over the declared spans; ties break to the
    # earliest phase_seq so the answer is stable rather than dependent on dict order.
    tally: dict[str, int] = {}
    for row in rows:
        best = max(spans, key=lambda s: (row[s["key"]], -s["phase_seq"]))
        key = best["label"] if row[best["key"]] > 0 else "unattributed"
        tally[key] = tally.get(key, 0) + 1

    # Largest contributor first, ties broken by phase_seq so the order is deterministic
    # rather than dependent on which order the rows happened to arrive in.
    seq = {s["label"]: s["phase_seq"] for s in spans}
    out = [" BREAKDOWN"]
    for label in sorted(tally, key=lambda k: (-tally[k], seq.get(k, 99))):
        if label != "unattributed":
            out.append(f"    {label:<28} {tally[label]}")
    # Printed even when zero, on purpose: an attribution that silently drops the orders it
    # could not explain is an attribution nobody can audit.
    out.append(f"    {'unattributed':<28} {tally.get('unattributed', 0)}")
    top = max((k for k in tally if k != "unattributed"), key=lambda k: tally[k])
    out += ["", f" Largest contributor: {top} ({tally[top]} of {len(rows)} late orders)."]

    # The finding that makes the case, and it is the opposite of the obvious suspect.
    under = [r for r in rows if r["transportation"] < 0]
    if under:
        out += ["", f" Transportation ran UNDER its standard on {len(under)} of "
                f"{len(rows)} late orders.",
                " Logistics was the obvious suspect and logistics was faster than",
                " planned. Without per-phase baselines you get a plausible, confident,",
                " wrong story -- and you go optimize the wrong department."]

    if qi.compare_to:
        # The prior window is compiled from the SAME metric definition, so the two numbers

```

```

# are comparable by construction. A comparison assembled from two hand-written
# queries is where "down 8.9 points" turns out to mean two different things.
prior = execute(con, qi, model, qi.compare_to)
out += ["", "  COMPARISON",
        f"    {'prior window':<26} {qi.compare_label} "
        f"({qi.compare_to[0]} to {qi.compare_to[1]}) on "
        f"{ans.metric['time_dimension']}"]
if prior.denominator:
    direction = "down" if prior.value > ans.value else "up"
    out += [f"    {'prior value':<26} {prior.value:.1f}% "
            f"({prior.numerator} / {prior.denominator})",
            f"    {'change':<26} {direction} "
            f"{abs(prior.value - ans.value):.1f}pp "
            f"({prior.value:.1f}% -> {ans.value:.1f}%)"]
else:
    out += [f"    {'prior value':<26} no eligible orders in the prior window",
            f"    {'change':<26} not computable"]
return "\n".join(out)

```

```

def _check_attribution_covers_the_late_orders(con, qi: QueryIntent, model: dict) -> None:
    """Fail loudly rather than attribute the wrong set.

```

`PHASE\_VARIANCE\_CSV` is embedded, while lateness is DERIVED from the order and leg tables. If those two ever disagree, the breakdown would silently describe a different population than the headline number. That is precisely the class of defect this whole project exists to prevent, so it is checked rather than assumed.

"""

```

window = f"""promised_delivery_date BETWEEN DATE '{qi.time_window[0]}'
            AND DATE '{qi.time_window[1]}'''

```

```

derived = {r[0] for r in con.execute(f"""
    SELECT order_id FROM fact_order_delivery
    WHERE is_eligible AND NOT is_on_time AND {window}""").fetchall()}

```

```

embedded = {r[0] for r in con.execute(f"""
    SELECT v.order_id FROM raw_phase_variance v
    JOIN fact_order_delivery f USING (order_id) WHERE {window}""").fetchall()}

```

```

if derived != embedded:
    raise AssertionError(
        f"attribution population mismatch -- derived late orders {sorted(derived)} "
        f"but PHASE_VARIANCE_CSV has {sorted(embedded)}"
    )

```

```

# =====
# 10. THE DEMONSTRATION
# =====

```

```

def _rule(title: str) -> str:
    return f"\n{' ' * 78}\n{title}\n{' ' * 78}"

```

```

def main() -> int:
    model = load_model()

```

```

con = build_warehouse()
failures = 0

# ---- 1. The governed answer -----
question = "What was OTD for India warehouses last month?"
print(_rule(f"1. THE GOVERNED ANSWER\n {question}"))
qi = resolve_offline(question, model)
print(f"\nThe resolver emitted this, and nothing else:\n{qi.to_json()}")
print(f"\nNote `grain` came from the metadata, not from the question.\n")
rc, out = answer(con, qi, model)
print(out)
failures += rc
governed = execute(con, qi, model)          # kept, so the gap below is measured

```

```

# ---- 2. What the same data yields without the grain declaration -----
print(_rule("2. THE SAME DATA, COUNTED AT DELIVERY GRAIN"))
naive = con.execute(f"""
    SELECT sum(CASE WHEN l.delivery_date <= o.promised_delivery_date
                      THEN 1 ELSE 0 END) AS on_time, count(*) AS legs
    FROM fact_order_delivery o
    JOIN raw_delivery_legs l USING (order_id)
    WHERE o.is_eligible
           AND o.promised_delivery_date BETWEEN DATE '2026-07-01' AND DATE '2026-07-31'
""").fetchone()
naive_pct = 100 * naive[0] / naive[1]
split = con.execute(f"""
    SELECT count(*) FROM fact_order_delivery
    WHERE is_eligible AND delivery_count > 1
           AND promised_delivery_date BETWEEN DATE '2026-07-01' AND DATE '2026-07-31'
""").fetchone()[0]
print(f"""

```

This is the query a text-to-SQL system writes. It is syntactically valid, semantically reasonable, and joins orders to legs without collapsing to one row per order:

```

{naive[0]} / {naive[1]} = {naive_pct:.4f}%    <-- reported as {naive_pct:.1f}%

```

{split} orders in this window shipped in more than one leg, so {split} orders voted twice. The gap is {naive_pct - governed.value:.2f} points -- small enough to look like rounding, and it errs UPWARD.

A wrong number that flatters you is the one that survives to the board deck."""")

```

# ---- 3. That query, refused -----
print(_rule("3. THAT QUERY, REFUSED BY NAME -- BEFORE ANY SQL EXISTS"))
print()
rc, out = answer(con, QueryIntent(
    intent_type="descriptive", metric="on_time_delivery_pct",
    grain="delivery",          # <-- the only change
    time_window=("2026-07-01", "2026-07-31"), time_label="July 2026",
    filters=(Filter("region", "=", "IN"),),
), model)
print(out)
failures += (rc != 1)

```

```

# ---- 4. Why it dropped -----

```

```

question = "Why did on-time delivery drop for India warehouses last month?"
print(_rule(f"4. THE QUESTION BI TOOLS CANNOT TAKE\n    {question}"))
print()
rc, out = answer(con, resolve_offline(question, model), model)
print(out)
failures += rc

# ---- 5. Three more gates -----
print(_rule("5. FOUR MORE CLASSES OF WRONG QUESTION, EACH REFUSED"))
base = dict(intent_type="descriptive", metric="on_time_delivery_pct", grain="order",
            time_window=("2026-07-01", "2026-07-31"))
for why, qi in [
    ("a filter value outside the declared allowed set",
     QueryIntent(**base, filters=(Filter("region", "=", "MARS"),)),
    ("a metric that exists but was never approved",
     QueryIntent(**base, "metric": "production_cycle_time_days")),
    ("an inverted time window",
     QueryIntent(**base, "time_window": ("2026-07-31", "2026-07-01"))),
    ("a dimension reachable only through a one-to-many join",
     QueryIntent(**base, dimensions=("delivery_id",))),
]:
    rc, out = answer(con, qi, model)
    gates = ", ".join(g for g, _ in validate(qi, model).failures)
    print(f"\n-- {why}\n    REFUSED by {gates}")
    failures += (rc != 1)

print(_rule("SUMMARY"))
print(f"""

```

The model chose one thing: which governed metric the question meant. Everything that had to be exactly right -- grain, exclusions, the join, the arithmetic, the window -- was compiled from metadata by code, the same way every time.

Two users asking the same question in different words get the same number, because the number was never in the model's hands.

```

{'OK -- every demonstration behaved as documented.' if not failures
 else f'FAILED -- {failures} demonstration(s) did not behave as documented.'}""")
    return 1 if failures else 0

```

```

if __name__ == "__main__":
    sys.exit(main())

```